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Summary

This paper introduces a novel approach to constructing prediction intervals by focusing on the concept of width-optimality, a departure from traditional methods that primarily emphasize coverage probability. The core contribution lies in the development of width-sketching algorithms,

designed to produce prediction intervals that not only maintain a desired coverage rate but also minimize the interval width. This is achieved through a novel penalty term that explicitly controls the width of the intervals. The authors argue that existing depth-sketching algorithms, while effective in ensuring coverage, often produce intervals that are unnecessarily wide, thus motivating the need for a width-centric approach. The paper presents two specific width-sketching algorithms, providing theoretical guarantees for their performance under certain conditions. These algorithms are based on the idea of using a penalty term to control the width of the prediction intervals. The theoretical analysis relies on concentration inequalities and other tools from probability theory. The paper also includes an experimental section where the proposed algorithms are evaluated on synthetic univariate and multivariate datasets. The results are compared against existing methods, such as the empirical quantile algorithm and a depth-sketching algorithm, demonstrating the effectiveness of the proposed approach in achieving better width-optimality. The authors also provide a discussion of the limitations of their approach, highlighting the assumptions that need to be satisfied for the algorithms to work well. Overall, the paper makes a significant contribution to the field of conformal prediction by introducing a new notion of optimality and developing algorithms that achieve this optimality. The theoretical guarantees and experimental results provide strong evidence for the effectiveness of the proposed approach, although there are areas where the paper could be improved, such as the clarity of the writing and the scope of the experimental validation.

✓ Strengths

I find the paper's introduction of width-optimality as a distinct goal in conformal prediction to be a significant and novel contribution. The authors have successfully identified a gap in existing methods, which often prioritize coverage at the expense of interval width. By explicitly addressing this issue, the paper offers a fresh perspective on the design of prediction intervals. The development of width-sketching algorithms, which incorporate a penalty term to control interval width, is a technical innovation that demonstrates a clear understanding of the problem. The theoretical analysis, while not without its limitations, provides a solid foundation for the proposed algorithms. The authors have provided rigorous proofs for their claims, demonstrating that their algorithms achieve the desired width-optimality under certain assumptions. The experimental results, although limited in scope, provide initial evidence that the proposed algorithms outperform existing methods in terms of width-optimality. The comparison against the empirical quantile algorithm and a depth-sketching algorithm provides a clear picture of the advantages of the proposed approach. The paper also includes a discussion of the limitations of their approach, highlighting the assumptions that need to be satisfied for the algorithms to work well. This demonstrates a balanced and critical approach to the research. The authors have also made an effort to connect their work to existing literature, citing relevant papers and building upon previous research. This helps to contextualize their contribution and demonstrate its significance. The paper is generally well-written and easy to follow, and the authors provide a clear motivation for their work. The mathematical formulations are presented in a clear and concise manner, making it easier for readers to understand the technical details. Overall, the paper presents a compelling argument for the importance of width-optimality in conformal prediction and provides a promising approach for achieving this goal.

✗ Weaknesses

While the paper presents a novel approach to width-optimality, several weaknesses need to be addressed. Firstly, the paper lacks a detailed comparison of the proposed width-sketching algorithm with existing methods, particularly depth-sketching algorithms, across key dimensions such as computational complexity, sample efficiency, and the tightness of the resulting confidence intervals. While the paper does compare against a specific depth-sketching algorithm by Angelopoulos et al. (2024), it does not provide a structured, quantitative comparison across these key dimensions. For instance, the paper mentions the computational complexity qualitatively as "polynomial complexity" but does not provide a detailed analysis in terms of the number of samples, feature dimension, and confidence level. This lack of a detailed comparison makes it difficult to fully understand the practical implications of the proposed method and its trade-offs compared to existing approaches. The absence of a comparative table summarizing these aspects further hinders a clear understanding of the advantages and disadvantages of each method. Secondly, the experimental section is limited in scope, relying solely on synthetic datasets. While the use of synthetic data is a common starting point, it is crucial to evaluate the performance of the proposed algorithms on real-world datasets to assess their robustness and generalizability. The current experiments, using only synthetic univariate and multivariate data, do not provide sufficient evidence to support the claim that the proposed algorithms are effective in practical scenarios. Furthermore, the experimental evaluation lacks comparisons against more established baselines in the field of conformal prediction, such as split conformal prediction or full conformal prediction. This omission

makes it difficult to assess the proposed algorithm's performance relative to existing state-of-the-art techniques. The paper also lacks explicit implementation details, such as pseudocode, which would make it easier for other researchers to reproduce the results and apply the proposed algorithm to their own problems. While the mathematical formulations are provided, translating them directly into code requires further steps. The paper mentions the use of the L1 norm as the width measure but doesn't detail other possible choices or their implementation. The paper also mentions the tolerance parameter δ but doesn't provide detailed guidelines for selecting other potential parameters. The computational complexity is discussed qualitatively, but a detailed analysis is missing. Finally, the paper suffers from numerous typographical errors, particularly in the mathematical expressions. These errors not only make the paper difficult to read but also raise concerns about the correctness of the results. For example, in Definition 2.2, the second condition is written as $\Pr(C(x^*) = \{u(x^*)\}) > \delta$ twice, which is clearly a typo. Similar repetitions occur in the other examples provided by the reviewer. These errors significantly hinder the readability and credibility of the paper. The writing is also unclear in some places, and the notation is not always well explained, making it difficult to follow the logic of the paper. The lack of a dedicated "Limitations" section also makes it difficult to assess the robustness and applicability of the proposed approach. While the paper does discuss limitations implicitly and through theorems, a dedicated section would provide a more balanced perspective. The confidence level for these weaknesses is high, as they are directly observable from the paper content and have been independently validated.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. Firstly, the authors should include a comprehensive comparative analysis of the proposed width-sketching algorithm with existing depth-sketching methods. This analysis should be presented in a table format, explicitly comparing the algorithms across key dimensions such as computational complexity, sample efficiency, and the tightness of the resulting confidence intervals. For each method, the authors should provide a concise explanation of the underlying mechanism that leads to its performance characteristics in each of these dimensions. For example, the authors could discuss how the specific choices of sketching dimensions or the use of different distance metrics affect the computational and sample complexity of the algorithms. This would provide a more concrete understanding of the trade-offs between different approaches and help readers to choose the most appropriate method for their specific needs. Secondly, the experimental section should be significantly expanded to include a more diverse set of datasets, including real-world datasets with varying characteristics, such as different data distributions, dimensionality, and sample sizes. This would help to assess the robustness and generalizability of the proposed method. Furthermore, the experimental evaluation should include a comparison against more established baselines in the field of conformal prediction, such as the split conformal method or the full conformal method. This would provide a more comprehensive evaluation of the proposed algorithm's performance relative to existing state-of-the-art techniques. The authors should also consider reporting the computational time required for each method, which would be useful for practical applications. Thirdly, the authors should provide a detailed description of the implementation details of the proposed algorithm. This should include pseudocode for the algorithm, a discussion of the specific choices of parameters that need to be made, and guidelines for parameter selection. The authors should also discuss the computational complexity of the algorithm and its scalability with the size of the dataset. This would make it easier for other researchers to reproduce the results and to apply the proposed algorithm to their own problems. Furthermore, the authors should consider providing an open-source implementation of the algorithm to facilitate its adoption by the research community. The discussion of implementation details should also include a discussion of the practical challenges that might arise during implementation, such as numerical stability issues or the need for specific data preprocessing steps. Fourthly, the authors should carefully proofread the paper to eliminate all typographical errors, particularly in the mathematical expressions. The notation should be consistently defined, and the logic of the paper should be made easier to follow. The authors should also consider adding a dedicated "Limitations" section to the paper. This section should include a discussion of the assumptions that need to be satisfied for the algorithm to work well, as well as any known limitations or potential failure modes. The authors should also discuss the robustness of the algorithm to violations of these assumptions. This would provide a more balanced evaluation of the proposed approach and help readers to understand its applicability and limitations. Finally, the authors should consider adding a more detailed discussion of the related work, highlighting the novelty and contributions of their approach. By addressing these issues, the authors can significantly improve the quality and impact of their paper.

Questions

Based on my analysis, I have several questions that I believe would be beneficial for the authors to address. Firstly, how does the proposed width-sketching algorithm compare to existing depth-sketching algorithms in terms of computational complexity and sample efficiency, specifically when considering varying data distributions, dimensionality, and sample sizes? While the paper provides some qualitative discussion, a more detailed analysis is needed to fully understand the practical implications of the proposed method. Secondly, what are the specific limitations of the proposed width-sketching algorithm, and under what conditions might it fail or perform poorly? The paper mentions some assumptions, but a more detailed discussion of the potential failure modes and the sensitivity of the algorithm to violations of these assumptions would be valuable. Thirdly, can the proposed width-sketching algorithm be extended to other types of prediction problems, such as regression or time series forecasting? The paper primarily focuses on classification problems, but exploring the potential for extending the approach to other domains would be a valuable direction for future research. Fourthly, what are the specific choices of parameters that need to be made when implementing the proposed algorithm, and how should these parameters be tuned in practice? The paper mentions the tolerance parameter δ but does not provide detailed guidelines for selecting other potential parameters. A more detailed discussion of parameter selection would be helpful for practitioners. Finally, what are the potential avenues for future research building upon the proposed width-sketching approach? The paper introduces a novel concept, and exploring its potential for further development would be beneficial for the research community. These questions target core methodological choices and seek clarification of critical assumptions, aiming to provide a more comprehensive understanding of the proposed approach.

 Scores**Soundness: 2.5****Presentation: 2.5****Contribution: 2.5****Rating: 4.75** AI Review from ZGCA will be automatically processed **Summary**

The paper proposes a new framework for uncertainty quantification centered on bounding the width of predictive sets uniformly over inputs, introducing "width-sketching algorithms." It argues that prior depth-focused methods (e.g., conformal risk control and depth-sketching algorithms) do not control width uniformly, and develops a penalty-based mechanism (a "width-sketching penalty" λ) and a reparameterization strategy to enforce width control "in expectation with high probability." The paper also attempts to formalize width-optimality and mean-width optimality, studies the relation between width and depth under convexity assumptions, gives counterexamples suggesting depth control does not imply width control, and sketches applications to conformal prediction with Bayesian quadrature.

✓ Strengths

- Motivates an important and underexplored objective: uniform width control as a complement to depth/coverage control (Section 1, Definition 1.1).
- Attempts to formalize a novel class of algorithms (width-sketching) distinct from depth-sketching and to articulate their complementarity (Sections 1 and 3.2).
- Identifies limitations of depth-sketching algorithms with respect to width (e.g., Theorem 1, Theorems 2–3), highlighting a potentially valuable research direction.
- Introduces a penalty-based mechanism (λ) and a reparameterization scheme (Section 4.1.4) intended to yield controllable width.
- Connects to Bayesian quadrature as a potential application domain (Abstract; Sections 2 and 5), which could be impactful if worked out rigorously.

✗ Weaknesses

- Core definitions are unclear or mathematically flawed. Definition 2.2 blends expectations, suprema, and conditioning in a way that is hard to interpret, and it is unclear what uniformity means operationally. Several objects (e.g., $C(x) \in \mathbb{R}$ or $\mathbb{R}^+ \times 2^{\mathbb{R}}$, intervals vs. sets) are conflated, and the role of $u(x)$, $g(x)$, and $v(x)$ is not consistently maintained (Sections 2 and 4).
- The definition of width-optimality in Equation (18) appears ill-posed: the denominator subtracts an expectation from itself, yielding division by zero. This undermines the central optimality notion.
- Multiple theoretical statements are either nonstandard or self-contradictory, e.g., Theorem 1 claims width in expectation is "greater than $O(\sqrt{n})$ ", which is not a meaningful asymptotic statement; Theorem 2 uses "controls the width" while simultaneously asserting an expectation exceeding n ; Theorems 5–7 contain claims like "at least four times smaller than the optimal width" (a contradiction in terms).
- The constructions defining empirical depth/width sketches (e.g., Equation (7) and (9)) are opaque: intersections of intervals formed from quantiles are not well-defined as written, s is chosen so that $C(n, s) = 1$ (nonsensical), and the role of the tree leaves π_s is unclear. The conditioning in (9) and probabilities over sets of leaves are not standard and are insufficiently justified.
- Key penalties and bounds (e.g., $\lambda_n(\delta)$ in (16) and (21)) are introduced without derivations or references that match the proposed form, and the guarantees in Theorem 10 are stated twice identically (Equations (22) and (23)), without proof details connecting the penalty to width control.
- Assumptions (convexity, bounded moments, i.i.d.) are moved between sections and reused inconsistently. Assumption 3.2 introduces unusual notation and constraints that are difficult to parse and verify. The relationship between Assumptions 3.4/3.5 and Theorem 4 is unclear, and (6) as written appears tautological or ill-typed.
- No empirical evaluation, no algorithm pseudocode, and no reproducible setup. The practical benefits are not demonstrated on synthetic or real datasets, unlike standard practice in conformal prediction and risk control.
- Exposition is frequently confusing with overloaded notation (x , x^* , $v(x)$), typos (e.g., 'Donut problem in expectation'), contradictory claims, repeated or malformed equations, and undefined terms (e.g., 'disjoint concourse').

? Questions

- Definition 2.2: Please precisely formalize the width-sketching guarantee. What is the exact notion of uniformity over x ? What is the measurable event being controlled, and with respect to which randomness (training data, algorithmic randomness, test point)?

- Equation (18): The denominator appears to subtract the same term from itself, leading to division by zero. Could you restate a correct and meaningful definition of width-optimality and mean-width optimality? How does it connect to classical notions such as minimizing expected set size subject to coverage constraints?
- Equations (7)–(9): Please provide a rigorous construction of $M_s(x, S)$. What are $z_{\{ij\}}$, how are the intervals $[a + \sigma(x) z_{\{ij\}}]$ defined (endpoints?), and how does the intersection yield an interval? What is the 's-deep tree' and the role of π_s ? A formal pseudocode or set-theoretic description is needed.
- The penalty $\lambda_n(\delta)$ in (16)/(21): What is the derivation and guarantee behind these expressions? Are they motivated by specific concentration inequalities (e.g., Hoeffding, Bentkus, DKW)? Please provide proofs or references that match the stated forms.
- Theorem 10: Please provide a complete proof showing how Definition 4.3 leads to the width bounds in (22)–(23), and clarify why (22) and (23) are stated twice identically. Under what assumptions on convexity, moments, and calibration does this result hold?
- Counterexamples (Theorems 1–3, 5–7): Can you provide precise constructions with standard probability space definitions and correct asymptotic statements (avoid 'greater than $O(\sqrt{n})$ ')? Please fix inconsistent phrasing such as 'at least four times smaller than the optimal width.'
- Relationship to depth-sketching (Section 3): Under convexity, which formal result connects depth control to width control? Can you state and prove a proposition that bounds expected width as a function of depth when the support is convex and bounded?
- Application to conformal prediction with Bayesian quadrature: Can you provide a concrete algorithm (pseudocode), a specification of model/data, and experiments demonstrating width control vs. baselines (e.g., split conformal, RCPS), ideally including ablations of the penalty λ ?
- Notation/parameterization: Please standardize the use of $u(x)$, $v(x)$, $g(x)$, and $C(x)$, and the domain/codomain of $C(x)$. If $C(x)$ is an interval $[l(x), r(x)]$, define width as $r(x) - l(x)$ and state expectations clearly. Avoid mixing intervals and sets in $\mathbb{R}^d$ without rigorous measure-theoretic definitions.

Limitations

- The framework currently relies on strong and sometimes unclear assumptions (e.g., convexity of $u(\cdot)$ ranges, bounded moment assumptions) to connect depth to width; without these, meaningful uniform width guarantees may be impossible for small δ (Section 2 mentions impossibility when $\delta < 1/(2d)$).
- A penalty-based approach may trade off depth/coverage for narrower width; careful calibration is needed to avoid undercoverage. The current manuscript does not quantify these trade-offs empirically.
- Width-optimality is ill-posed as stated; even with a corrected definition, minimization of expected width subject to conditional coverage is known to be challenging and may require distributional or structural assumptions.
- Scalability and computational complexity are not addressed; the constructions involving intersections and trees (π_s) appear potentially expensive.
- Potential negative societal impacts are minimal in this paper's form, but deploying width-minimizing prediction sets in high-stakes settings could inadvertently reduce coverage for minority subgroups if not properly conditioned. Addressing conditional coverage and fairness-aware calibration would mitigate this.

Image Evaluation

Cross-Modal Consistency: 14/50

Textual Logical Soundness: 8/30

Visual Aesthetics & Clarity: 6/20

Overall Score: 28/100

Detailed Evaluation (≤500 words):

1. Cross-Modal Consistency

- Major 1: Missing referenced table; the text relies on a non-existent summary, blocking verification. Evidence: “see Tab. 1 for a summary”
- Major 2: Type/range mismatch for C . Definition and prose disagree (tuple vs scalar). Evidence: “ $C: \dots \rightarrow \mathbb{R}^+ \times 2\mathbb{R}$ ” and “ $C_{\{A_n\}}: \mathbb{R}^d \times (0,1) \rightarrow \mathbb{R}$ ”
- Major 3: Semantic conflation of coverage δ and depth α across sections. Evidence: “increase the desired depth $\alpha(\delta) := \min(\delta, \alpha_i)$ ”
- Minor 1: Ambiguous set notation (“ $2\mathbb{R}$ ”, “ $2^{\mathbb{R}}$ ”) causing uncertainty about codomain. Evidence: “ $C(x) \subseteq 2\mathbb{R}$ ”
- Minor 2: Symbols introduced without consistent legends (e.g., π_s , S_+). Evidence: “ π_s is a set of s leaves of the tree”

2. Text Logic

- Major 1: Contradictory optimality statements (claims “smaller than optimal”). Evidence: “width... is at least four times smaller than the optimal width”
- Major 2: Ill-formed inequality (same quantity both sides; extraneous bracket). Evidence: Eq. (6)
- Major 3: Unsubstantiated extremal growth claim without construction or conditions. Evidence: “width... $> n$ for all $n > 0$ ”
- Minor 1: Undefined/unclear constructs hinder reasoning chain. Evidence: “is Donut problem in expectation”
- Minor 2: Frequent notation flips (depth vs width, $1-\alpha$ vs δ) impede clarity. Evidence: “They provide complementary forms of coverage... we take $\delta_+ := \delta - \lambda$ ”

3. Figure Quality

- Major 1: No figures/tables provided despite explicit references; readers cannot cross-check claims. Evidence: “see Tab. 1 for a summary”
- Minor 1: Equations act as core visuals but lack clear legends/units/assumptions (e.g., S_+ , π_s). Evidence: “ $S_+ := \{u(x): u(x) \in [\delta/(1-\alpha), 1]\}$ ”

Key strengths:

- Ambitious attempt to formalize “width-sketching” and relate it to depth-based conformal methods.
- Highlights a practical need (controlling expected width) and proposes a penalty-based approach.

Key weaknesses:

- Severe notation and definition inconsistencies (domain/codomain of C , use of $2\mathbb{R}$ vs $2^{\mathbb{R}}$).
- Multiple logically incorrect or contradictory theorems (e.g., “smaller than optimal”).
- Several ill-formed equations and undefined symbols; unclear probabilistic setup (nested expectations, trees).
- Missing referenced table; no diagrams to clarify the framework; hard-to-follow proofs and assumptions.
- Conflation of depth (α) and coverage/quantile (δ) across sections undermines the argument chain.

Recommendations:

- Unify notation for C , g , width measure W , and domains; fix $2\mathbb{R} \rightarrow 2^{\mathbb{R}}$ where intended.
- Correct Theorems 5–7 statements and provide precise constructions/proofs.
- Define all symbols (S_+ , π_s , L_s , M_s) once, with a consistent glossary.
- Add a summary table (the referenced Tab. 1) and a schematic pipeline figure to clarify the algorithm/assumption flow.
- Separate and consistently use δ (coverage) and α (depth/tolerance) across all definitions and bounds.

 ScoresOriginality: **3**Quality: **1**Clarity: **1**Significance: **2**Soundness: **1**Presentation: **1**Contribution: **2****Rating: 2** AI Review from SafeReviewer will be automatically processed **Summary**

This paper introduces the concept of width-sketching algorithms, which aim to construct prediction intervals with controlled width while maintaining a desired coverage probability. The authors identify a gap in existing depth-sketching methods, which, while ensuring coverage, often produce intervals with unbounded or excessively wide intervals. To address this, they propose a novel framework that incorporates a width-sketching penalty, allowing for a more balanced trade-off between coverage and interval width. The core idea is to modify existing depth-sketching algorithms by adding a penalty term that explicitly controls the width of the resulting intervals. The paper presents theoretical results demonstrating that their proposed width-sketching algorithm can achieve a desired coverage probability while bounding the expected width of the prediction intervals. They also introduce the concepts of width-optimality and mean-width optimality, providing a theoretical framework for evaluating the efficiency of interval estimation. The authors demonstrate the application of their framework to conformal prediction with Bayesian

quadrature, showcasing its practical relevance. The paper's main contribution lies in the formalization of width-sketching as a distinct problem and the development of a corresponding algorithmic framework. The theoretical results provide a foundation for understanding the trade-offs between coverage and width, and the experimental results, while limited, demonstrate the potential of the proposed approach. However, the paper's presentation and clarity need significant improvement to make the core ideas more accessible and understandable. The lack of explicit algorithmic descriptions, unclear definitions, and insufficient experimental details hinder the paper's overall impact and practical applicability. Despite these limitations, the paper tackles an important problem in uncertainty quantification and provides a valuable contribution to the field by introducing the concept of width-sketching and its potential for improving the efficiency of prediction intervals.

✓ Strengths

The paper's primary strength lies in its introduction of the concept of width-sketching algorithms, which addresses a critical gap in existing methods for constructing prediction intervals. I find the motivation for this new approach compelling, as it directly tackles the issue of unbounded or excessively wide intervals produced by depth-sketching methods. The authors correctly identify that while depth-sketching algorithms ensure coverage, they often fail to control the width of the resulting intervals, leading to impractical predictions. The proposed width-sketching framework, which incorporates a penalty term to explicitly control interval width, represents a novel contribution to the field. I appreciate the theoretical results that demonstrate the ability of the proposed algorithm to achieve a desired coverage probability while bounding the expected width of the prediction intervals. The introduction of width-optimality and mean-width optimality provides a valuable theoretical framework for evaluating the efficiency of interval estimation. Furthermore, the application of the proposed framework to conformal prediction with Bayesian quadrature highlights its practical relevance and potential for real-world applications. This demonstrates the authors' ability to connect theoretical concepts with practical problems, which is a significant strength. The paper also builds upon existing work in conformal prediction and depth-sketching, which shows a good understanding of the current literature. The authors identify the limitations of existing methods and propose a novel approach to address these limitations, which is a valuable contribution to the field. Overall, the paper's strengths lie in its novel concept of width-sketching, its theoretical results, and its practical application to conformal prediction.

✗ Weaknesses

After a thorough examination, I've identified several significant weaknesses that hinder the paper's clarity and impact. Firstly, the paper suffers from a lack of clarity in its definitions and notation. For instance, the definition of a width-sketching algorithm (Definition 2.2) is difficult to parse, and the role of the index 'n' is not explicitly stated within the definition itself, although it is implied to be the index of the algorithm in a sequence. The paper also uses the term 'width' without a clear definition until later, and even then, the connection to the general case isn't immediately obvious. The notation $C(x^*)$ is used before its definition, and the meaning of $M(u(x))$ is unclear. This lack of clarity makes it challenging to understand the core concepts being presented. Secondly, the paper lacks explicit algorithmic descriptions. While the authors describe the components of their proposed algorithm, such as the width-sketching penalty, they do not provide a step-by-step procedure or pseudocode. This makes it difficult to understand how the algorithm is implemented in practice. For example, the description of the empirical width sketch is vague, and the connection between the empirical width sketch and the L1 norm is not immediately clear. The paper also lacks a clear explanation of how the width-sketching penalty is incorporated into the algorithm. Thirdly, the paper's experimental section is limited and lacks crucial details. The experiments do not include real-world datasets, and the description of the experimental setup is insufficient. The paper does not clearly define the metrics used, and the results are not thoroughly analyzed. The choice of baselines is not well-justified, and the paper does not discuss the limitations of the proposed method in the context of the experiments. The paper also lacks a clear explanation of the practical implications of the theoretical results. While the theoretical results are interesting, the paper does not clearly explain how these results translate into practical benefits. The paper also does not discuss the computational complexity of the proposed algorithm or compare it to existing methods. Finally, the paper's writing style is not always clear, and the presentation of the results is not always well-organized. The paper uses a lot of technical jargon, which makes it difficult for a non-expert to understand. The paper also lacks a clear narrative, and the results are not always presented in a logical order. The lack of a clear connection between the theoretical results and the experimental results also makes it difficult to assess the overall significance of the paper. The paper also does not adequately address the limitations of the proposed method or

discuss potential avenues for future research. The lack of a clear explanation of the practical implications of the theoretical results also makes it difficult to assess the overall significance of the paper. The paper also does not discuss the computational complexity of the proposed algorithm or compare it to existing methods. These weaknesses significantly impact the paper's overall clarity, impact, and practical applicability. The lack of clear definitions, explicit algorithms, detailed experiments, and a clear explanation of the practical implications of the theoretical results makes it difficult to fully appreciate the contributions of the paper. The writing style and organization also need significant improvement to make the paper more accessible and understandable.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the authors should significantly enhance the clarity of their definitions and notation. They should explicitly define all terms before their first use, including 'width,' $M(u(x))$, and the role of 'n' in Definition 2.2. They should also provide a more intuitive explanation of the width-sketching algorithm, perhaps with a simple example to illustrate its behavior. Second, the authors should provide explicit algorithmic descriptions, including pseudocode, for their proposed methods. This would greatly improve the reproducibility of their results and make it easier for others to understand and build upon their work. For example, they should provide a clear step-by-step description of how the empirical width sketch is constructed and how the width-sketching penalty is incorporated into the algorithm. Third, the authors should expand their experimental section to include real-world datasets and provide more detailed descriptions of their experimental setup. They should clearly define the metrics used, provide a thorough analysis of their results, and justify their choice of baselines. They should also discuss the limitations of their proposed method in the context of the experiments and compare its computational complexity to existing methods. For example, they could use standard benchmark datasets for prediction interval construction and compare their method to established techniques. Fourth, the authors should provide a more detailed explanation of the practical implications of their theoretical results. They should clearly explain how their theoretical findings translate into practical benefits and provide concrete examples to illustrate their points. They should also discuss the computational complexity of their proposed algorithm and compare it to existing methods. For example, they could discuss the trade-offs between coverage and width and provide guidance on how to choose the parameters of their algorithm in practice. Fifth, the authors should improve the overall writing style and organization of the paper. They should use less technical jargon, provide a clear narrative, and present their results in a logical order. They should also ensure that all figures and tables are properly referenced in the text and that the captions are informative. For example, they should explicitly refer to Figure 1 and Table 1 in the text and provide clear explanations of their contents. Finally, the authors should address the limitations of their proposed method and discuss potential avenues for future research. They should also provide a more detailed discussion of the related work and clearly articulate the novelty of their contributions. By addressing these points, the authors can significantly improve the clarity, impact, and practical applicability of their paper.

Questions

After reviewing the paper, I have several questions that I believe are crucial for a deeper understanding of the proposed methods and their implications. First, regarding the definition of the width-sketching algorithm (Definition 2.2), what is the precise meaning of the condition involving $M(u(x))$? It is not clear what this term represents, and a more detailed explanation is needed. Second, in the context of the empirical width sketch, how is the L1 norm applied, and why is it considered the 'standard' measure of width in this context? A more detailed explanation of the connection between the empirical width sketch and the L1 norm is needed. Third, how does the proposed width-sketching penalty explicitly control the width of the resulting intervals, and what are the theoretical guarantees for this control? A more detailed explanation of the mechanism of the penalty term and its impact on the interval width is needed. Fourth, what is the computational complexity of the proposed algorithm, and how does it compare to existing methods? A more detailed analysis of the computational cost of the algorithm is needed. Fifth, what are the limitations of the proposed method, and under what conditions might it fail to perform well? A more detailed discussion of the limitations of the method is needed. Sixth, how does the choice of the parameter λ affect the trade-off between coverage and interval width, and what are the practical guidelines for choosing this parameter? A more detailed explanation of the role of this parameter is needed. Seventh, how does the proposed method compare to other existing methods for constructing prediction intervals, such as conformal prediction, in terms of

both performance and computational cost? A more detailed comparison with existing methods is needed. Finally, what are the potential avenues for future research based on the proposed width-sketching framework? A more detailed discussion of the potential future directions is needed. These questions target core methodological choices, critical assumptions, and key uncertainties that need to be addressed for a more complete understanding of the paper's contributions.

 Scores

Soundness: **2.0**

Presentation: **1.0**

Contribution: **1.67**

Rating: 3.0

Keywords

Extract Keywords

Click the button to extract keywords

Insights

Extract Insights

Click the button to extract insights